EXPERIMENT 27 : Implementation of Monkey Banana Problem Using Prolog

Aim:

To write and execute a Prolog program to implement the Monkey Banana Problem using state-space search.

Software Required:

- SWI-Prolog
- Windows Operating System

Theory:

The Monkey Banana Problem is a classical Artificial Intelligence problem. A monkey is in a room containing a box and bananas hanging from the ceiling. The monkey must walk towards the box, push it to the middle of the room, climb onto it and grab the bananas.

The problem is solved using state-space search, where each state represents the positions of the monkey and box and whether the monkey has climbed or obtained the bananas.

Program:

Save the file as exp27.pl in C:/AI Prolog/.

```
% Monkey Banana Problem

% State: state(Monkey, Box, Status, Banana)

% Monkey walks
move(state(door, window, onfloor, no),
    walk,
    state(window, window, onfloor, no)).

% Monkey pushes the box
move(state(window, window, onfloor, no),
    push,
    state(middle, middle, onfloor, no)).
```

```
% Monkey climbs the box
move(state(middle, middle, onfloor, no),
     climb,
     state(middle, middle, onbox, no)).

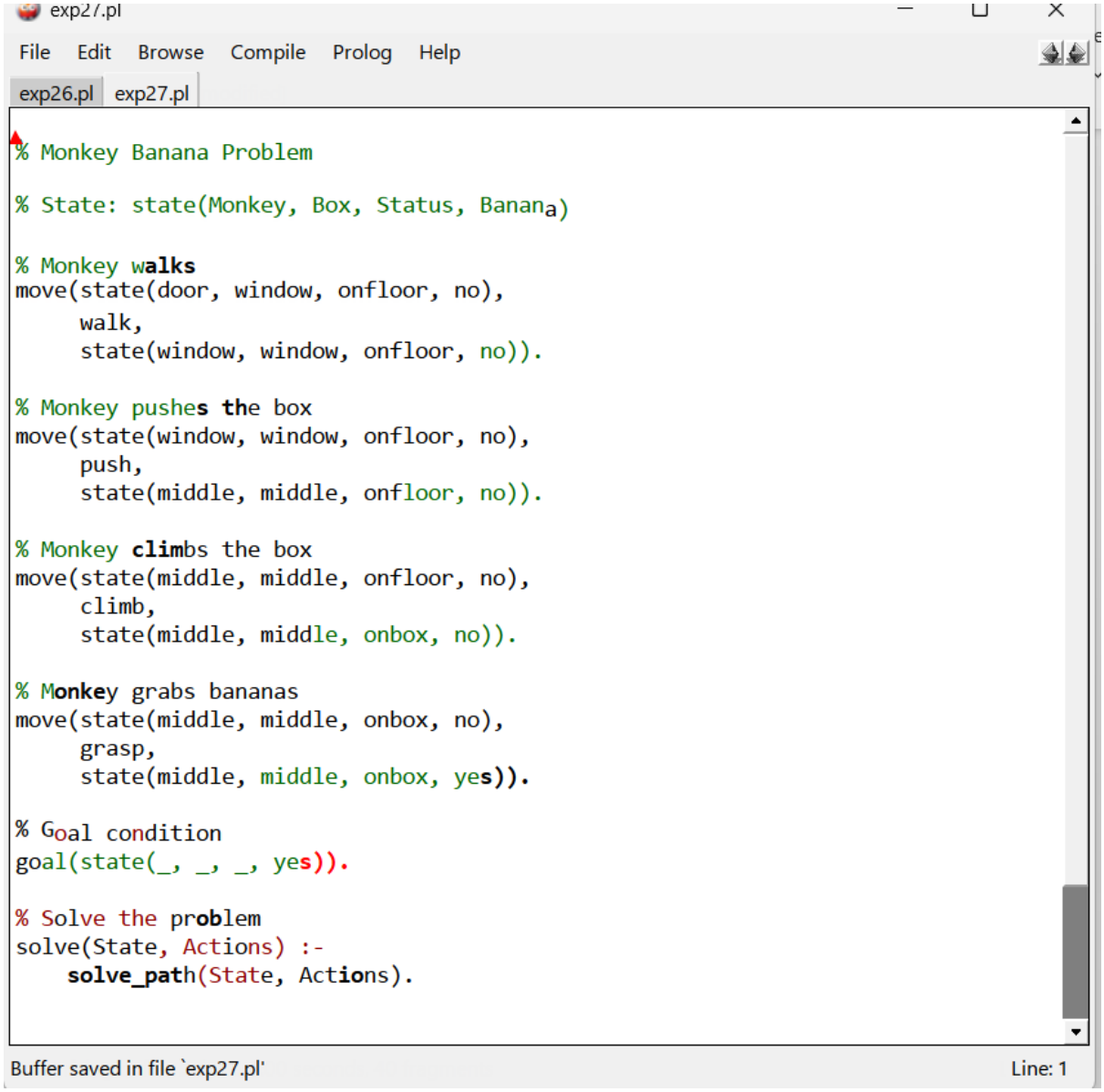
% Monkey grabs bananas
move(state(middle, middle, onbox, no),
     grasp,
     state(middle, middle, onbox, yes)).

% Goal condition
goal(state(_, _, _, yes)).

% Solve the problem
solve(State, Actions) :-
    solve_path(State, Actions).

solve_path(State, []) :-
    goal(State).

solve_path(State, [Action|Rest]) :-
    move(State, Action, NextState),
    solve_path(NextState, Rest).
```



Procedure:

1. Open SWI-Prolog.
2. Create a new file named exp27.pl.
3. Type the above Prolog program.
4. Save the file in C:/AI Prolog/.
5. Load the program using the following command:

?- consult('c:/ai prolog/exp27.pl').

6. Execute the following query:

?- solve(state(door, window, onfloor, no), Actions).

Output:

```
80 ?- consult('c:/ai prolog/exp27.pl').
true.

80 ?- solve(state(door, window, onfloor, no), Actions).
Actions = [walk, push, climb, grasp]
```

Algorithm:

1. Initialize the monkey and box positions.
2. Move the monkey from the door to the window.
3. Push the box from the window to the middle of the room.
4. Climb onto the box.
5. Grab the bananas.
6. Display the sequence of actions.

Result:

Thus, the Prolog program to implement the Monkey Banana Problem using state-space search was successfully written and executed using SW